

Notes: Chen, An, Wang & Yu (JF, 2023)
Attention Spillover in Asset Pricing

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Trading-software display and listing-code distance	4
3.2	Retail brokerage account data	4
3.3	Survey-linked trading data	4
3.4	Stock-level sample and LOCAL/RLOCAL	5
3.5	Returns, turnover, and control variables	5
4	Models and empirical specifications	5
4.1	Positive feedback trading at the investor level	5
4.2	Attention spillover by display distance	5
4.3	Panel regression for microfoundation	6
4.4	Survey mechanism regression	6
4.5	Extreme-return/placebo mechanism test	6
4.6	Portfolio tests of return predictability	6
4.7	Fama-MacBeth return and turnover regressions	6
4.8	Quasi-natural experiment around SME Board introduction	7
5	Identification strategy	7
5.1	Core source of quasi-random attention variation	7
5.2	Why own-return residualization matters	7
5.3	Microfoundation before pricing	7
5.4	Mechanism separation	7
5.5	Quasi-natural experiment	8
5.6	Remaining limitations	8
6	Tables and figures	8
6.1	Figure 1: Screen display of the trading software and Figure 2: Listing code and stock characteristics	8
6.2	Figure 3: Positive feedback trading and attention spillover and Table I: Positive Feedback Trading and Attention Spillover – Panel Regressions	9
6.3	Table II: The Primary Determinants of Positive Feedback Trading	11
6.4	Table III: Experienced Returns versus Observed Extreme Returns	11
6.5	Table IV: Summary Statistics and Table V: Single-Sorted Portfolio Return by RLOCAL	12

6.6	Table VI: Double-Sorted Portfolio Return by RLOCAL and Confounding Variables and Table VII: Fama-MacBeth Regressions	12
6.7	Table VIII: Fama-MacBeth Regressions for Placebo Tests and Table IX: Forecasting Turnover by RLOCAL	13
6.8	Figure 4: Comovement in Return and Turnover and Distance between Stocks and Table X: Comovement in Return and Turnover Difference-in-Difference	13
6.9	Table XI: Performance of RLOCAL Portfolios in the Post-SME Period and Figure 5: Cumulative Return of the Long-Short Portfolio Based on RLOCAL	14
6.10	Table XII: Cost of Arbitrage and RLOCAL Returns	15
7	Interpretive synthesis	15
8	Short summary	16
9	Conclusion	16
9.1	Core conclusion	16
9.2	Economic intuition	16
9.3	Incremental contribution revisited	16
9.4	What the paper does not claim	17
9.5	Final takeaway	17

1 Big picture

- **Question.** How does the interaction between investor overconfidence and limited attention affect trading, short-term prices, turnover, and comovement?
- **Setting.** The paper studies Chinese A-share stocks and exploits a trading-software display rule: stocks are displayed in the order of their listing codes. Stocks with nearby listing codes are mechanically shown close to one another on investors' screens.
- **Key behavioral mechanism.** Investors who experience positive returns become more willing to trade. At the same time, they are especially likely to notice stocks displayed near the stock they already hold. The interaction creates buying pressure in neighboring stocks after the investor's held stock performs well.
- **Main stock-level signal.** *LOCAL* is the value-weighted average return over the past two weeks of the 10 closest stocks on the screen display. *RLOCAL* residualizes *LOCAL* on the focal stock's own past two-week return to reduce reflection and short-term autocorrelation concerns.
- **Main return result.** Stocks with high *RLOCAL* earn higher next-week returns. The single-sorted long-short spread is about 8% per year and remains significant after age, industry, characteristic, CH4, and FF5 adjustments.
- **Reversal result.** Long-short returns based on *RLOCAL* are temporary and reverse in the long run, consistent with transitory price pressure rather than fundamentals.
- **Trading-volume result.** *LOCAL* also predicts turnover, especially small-buy turnover, which is consistent with retail buying pressure.
- **Comovement result.** Stocks that are closer on screen display have higher return and turnover comovement, while accounting fundamentals do not show the same distance gradient.
- **Quasi-natural experiment.** The introduction of the SME Board exogenously increases display distance for some stocks; comovement falls more for pairs whose display distance increases.
- **Economic takeaway.** Non-informational screen adjacency can shape attention, trading, returns, turnover, and comovement when combined with investors' tendency to trade more after good personal outcomes.

2 Incremental contribution of the paper

- **Interaction rather than one bias at a time.** Prior asset-pricing work studies limited attention or overconfidence separately. This paper shows that their interaction can generate a price effect that neither bias alone would produce in this setting.
- **Clean attention variation.** Attention is identified from listing-code proximity on trading screens. This is more plausibly exogenous than news, search, volume, or extreme returns, which may all contain fundamental information.
- **Micro-to-market chain.** The paper connects individual account-level behavior to aggregate stock-level pricing. It first documents positive-feedback trading and attention spillover in retail accounts, then builds *LOCAL*/*RLOCAL* to test stock returns and turnover.
- **Distinct from underreaction.** Many attention papers study underreaction to information. Here the mechanism is continued overreaction: investors buy visible neighbors after good experience, pushing prices up temporarily.
- **Behavioral mechanism measurement.** The investor survey distinguishes self-attribution, extrapolation, gambling preference, home bias, perceived information advantage, and other behavioral channels. Self-attribution is the strongest predictor of positive feedback trading.
- **Additional identification from a display shock.** The SME Board introduction provides a quasi-natural experiment that changes screen-display proximity without directly changing fundamentals.

- **Broader lesson.** Financial-market interfaces can affect equilibrium outcomes. Seemingly innocuous ordering rules in trading software can become economic primitives when investors have limited attention.

3 Data and measurement

3.1 Trading-software display and listing-code distance

The key institutional feature is that common Chinese trading platforms allow investors to search by ticker name or listing code, page up/down to adjacent listing codes, and view a list of nearby stocks ordered by listing code.

- **Distance.** Distance is the absolute difference in display rank between two stocks. In many account-level tests, distance bins are $(0, 5]$, $(5, 10]$, ..., $(45, 50]$.
- **Why distance is meaningful.** The screen interface makes nearby listing-code stocks easier to notice after an investor looks at a held stock.
- **Why distance is not mechanically fundamental.** Listing codes are related to listing date and board, but the paper shows limited relation between code order and industry, province, or market capitalization after board-by-board inspection.

Interpretation. The display rule is the empirical lever: the order of stocks on the screen creates quasi-random local visibility conditional on listing board, age, and standard stock characteristics.

3.2 Retail brokerage account data

- The main brokerage data contain daily trading and holdings records for 401,014 investors from January 2009 to September 2012.
- The investor-day-currently-held-stock unit is used to measure whether investors buy new stocks after holding a winning or losing position.
- Because the full Cartesian set is extremely large, some regression specifications randomly select 50,000 investors.
- A “winning” position is a held stock that has increased in value since the investor purchased it; a “losing” position has decreased in value.

Interpretation. This data set supplies the microfoundation. The paper can directly observe that investors buy more after gains and that their new purchases are skewed toward screen-neighboring stocks.

3.3 Survey-linked trading data

- The authors conduct a survey in October 2021 through a brokerage trading app.
- The final survey sample used in Table II contains 1,229 surveyed investors and their trading data from January 2020 to January 2022.
- The survey contains 12 questions designed to capture behavioral biases, including self-attribution, extrapolation, overplacement, gambling preference, realized utility, home bias, and perceived information advantage.

Interpretation. The survey is not used to price stocks directly. It is used to identify which psychological mechanisms predict the positive-feedback behavior that generates attention-induced buying pressure.

3.4 Stock-level sample and LOCAL/RLOCAL

- The stock-level pricing tests use Chinese A-shares on the Shanghai and Shenzhen stock exchanges from January 2002 to December 2019.
- For stock i at week t , $LOCAL_{i,t}$ is the value-weighted average return over the previous two weeks of the 10 closest stocks in screen-display order.
- $RLOCAL_{i,t}$ is the residual from the cross-sectional regression of $LOCAL_{i,t}$ on the focal stock's own previous two-week return.
- Portfolios are formed weekly by sorting stocks into RLOCAL quintiles and holding them for the next week.

$$LOCAL_{i,t} = \sum_{j \in N_i} w_{j,t} R_{j,t-2:t}, \quad RLOCAL_{i,t} = LOCAL_{i,t} - \hat{\alpha}_t - \hat{\beta}_t R_{i,t-2:t}.$$

Interpretation. LOCAL measures potential buying pressure coming from owners of neighboring stocks. RLOCAL removes the focal stock's own recent return so the result is not simply short-term return continuation of the focal stock.

3.5 Returns, turnover, and control variables

- Return tests use raw returns, age-adjusted returns, industry-adjusted returns, DGTW characteristic-adjusted returns, Chinese four-factor alphas, and Fama-French five-factor alphas.
- Control variables include size, beta, book-to-market, past 12-to-2 month returns, past 36-to-13 month returns, Amihud illiquidity, turnover, idiosyncratic volatility, maximum return, skewness, exchange, and age.
- Turnover tests distinguish total turnover, abnormal turnover, small-order turnover, and order imbalances.

Interpretation. These controls address standard anomalies and market microstructure channels. The key question is whether a non-fundamental screen-neighbor signal predicts future returns and turnover after controlling for known predictors.

4 Models and empirical specifications

4.1 Positive feedback trading at the investor level

The first micro test asks whether investors buy more after a winning position than after a losing position:

$$\text{Exp}(\#buy \mid win) - \text{Exp}(\#buy \mid lose) > 0.$$

Interpretation. If positive past experience makes investors overconfident or self-attributing, they should become more willing to trade. This generates the first ingredient in the paper's mechanism.

4.2 Attention spillover by display distance

The second micro test asks whether new purchases are more likely to be close in screen-display distance to the currently held stock:

$$\Pr(dist = x \mid buy, win) \quad \text{and} \quad \Pr(dist = x \mid buy, lose).$$

Interpretation. If investors are more likely to notice nearby stocks, purchase probability should decline with distance. Combining this with positive-feedback trading implies extra buying pressure in nearby stocks after the held stock performs well.

4.3 Panel regression for microfoundation

The regression in Table I estimates distance-bin purchase probabilities:

$$1_{(x,x+5]} = \alpha_{(x,x+5]} + \beta_{(x,x+5]} 1_{win} + controls + \varepsilon.$$

Interpretation. A positive β at short distances and a declining pattern across distance bins means that winning positions lead to more neighbor purchases, and the effect is strongest for the closest stocks.

4.4 Survey mechanism regression

The survey-linked regressions test whether behavioral biases amplify positive feedback trading:

$$1_{i,t+1}^{net\ buy} = \alpha + \beta 1_{i,t}^{win} + \sum_j \theta_j 1_{i,j}^{bias} + \sum_j \gamma_j 1_{i,t}^{win} \times 1_{i,j}^{bias} + \varepsilon_{i,t+1}.$$

Interpretation. The interaction coefficients show which behavioral traits make investors more likely to buy after winning positions. The strongest interactions are self-attribution variables, supporting a learning-to-be-overconfident channel.

4.5 Extreme-return/placebo mechanism test

For daily price-limit events, the paper estimates:

$$1_{(x,x+5]} = \alpha_{(x,x+5]} + \beta_{(x,x+5]}^{hold} 1_{hold} + \beta_{(x,x+5]}^{hit} 1_{hit} + \beta_{(x,x+5]}^{inter} 1_{hold} \times 1_{hit} + \varepsilon.$$

Interpretation. This pits the experienced-return channel against a pure attention-grabbing event. Holding a stock with an extreme positive return has much more predictive power for neighbor purchases than merely observing that a stock hits the upper price limit.

4.6 Portfolio tests of return predictability

Stocks are sorted into RLOCAL quintiles. The main long-short return is:

$$R_{t+1}^{P5-P1} = R_{t+1}^{High\ RLOCAL} - R_{t+1}^{Low\ RLOCAL}.$$

Interpretation. A positive spread means that stocks near recent winners in screen space earn higher future returns. The paper interprets this as temporary buying pressure rather than compensation for risk.

4.7 Fama-MacBeth return and turnover regressions

The paper estimates cross-sectional predictive regressions:

$$\begin{aligned} R_{i,t+1} &= \alpha_t + \beta_t LOCAL_{i,t} + \gamma'_t X_{i,t} + \varepsilon_{i,t+1}, \\ Turnover_{i,t+1} &= \alpha_t + \beta_t LOCAL_{i,t} + \gamma'_t X_{i,t} + \varepsilon_{i,t+1}. \end{aligned}$$

Interpretation. LOCAL should predict both returns and turnover if the mechanism is buying pressure. Turnover is especially useful because a return-only result could be confused with expected-return variation.

4.8 Quasi-natural experiment around SME Board introduction

The SME Board introduction changes display distance for some stock pairs. The DID framework compares changes in correlation for treated pairs whose display distance increases relative to pairs whose distance does not:

$$Corr_{pair,t} = \alpha + \delta(TreatedPair \times Post_t) + pairFE + timeFE + \varepsilon_{pair,t}.$$

Interpretation. A negative DID effect means that increasing screen distance reduces return/turnover comovement, which supports attention-based comovement rather than shared fundamentals.

5 Identification strategy

5.1 Core source of quasi-random attention variation

The paper’s central identification argument is that stock adjacency in the trading interface is determined by listing code, not by current fundamentals, news, or investor beliefs. Conditional on listing board, age, and controls, nearby code stocks should not systematically share information shocks.

5.2 Why own-return residualization matters

RLOCAL residualizes LOCAL on the focal stock’s own past two-week return. This addresses two concerns:

- **Reflection problem.** A focal stock’s own past return might attract attention to nearby stocks and then be mechanically reflected in LOCAL.
- **Return autocorrelation.** If the focal stock’s own return predicts its next return, LOCAL could proxy for the focal stock’s momentum or reversal.

5.3 Microfoundation before pricing

The paper does not start with return predictability alone. It first documents two necessary behavioral ingredients:

- Investors trade more after personally experienced gains.
- Conditional on buying, they disproportionately buy stocks close to current holdings in display order.

Interpretation. This strengthens identification because the asset-pricing signal is motivated by observed trading behavior rather than data-mined return predictability.

5.4 Mechanism separation

The paper distinguishes the interaction mechanism from alternatives by using:

- survey measures of behavioral bias;
- price-limit events as attention-grabbing placebo shocks;
- LOCAL variants based on turnover and volatility;
- distant-neighbor placebo variables;

- long-run reversal tests;
- cost-of-arbitrage cross-sectional tests.

Interpretation. These tests ask whether the effect is truly about the interaction of experienced positive returns and screen attention, rather than generic salience, news, return momentum, common fundamentals, or investor underreaction.

5.5 Quasi-natural experiment

The SME Board event exogenously changes some display distances. If attention spillover is real, correlations should decline for pairs moved farther apart. This is a sharper test because it uses a change in the display environment rather than cross-sectional proximity alone.

5.6 Remaining limitations

- The setting is the Chinese stock market, where retail participation and short-sale constraints are important.
- Listing code assignment is not fully random; therefore board, listing age, industry, and characteristic adjustments are important.
- The retail brokerage data do not cover the entire market, so the mapping from observed retail trades to aggregate price pressure is indirect.
- The paper identifies temporary price pressure, not necessarily welfare losses or the full general-equilibrium impact of trading-platform design.

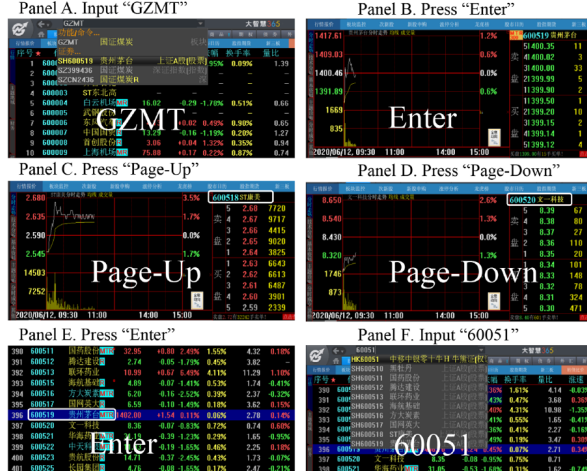
6 Tables and figures

6.1 Figure 1: Screen display of the trading software and Figure 2: Listing code and stock characteristics

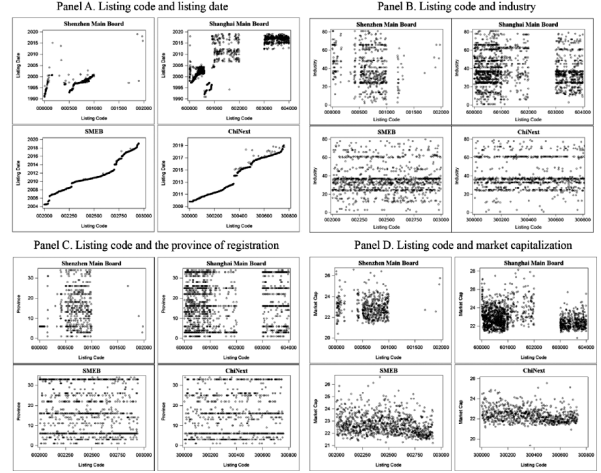
Read: Figure 1 illustrates how investors see nearby listing-code stocks when they search, press page up/down, or view the surrounding code list. Figure 2 shows that listing code is mechanically related to listing date on some boards, but less systematically related to industry, province, or market capitalization.

Interpretation. Figure 1 is the institutional foundation of the paper: attention can spill over mechanically through the trading interface. Figure 2 is a validity check: if code proximity simply captured industry, province, or size clusters, LOCAL would be harder to interpret as attention.

Economic message. The economic message is that a purely interface-driven ordering rule can create local attention neighborhoods that are plausibly unrelated to current fundamentals.



(a) Figure 1. Screen display of the trading software



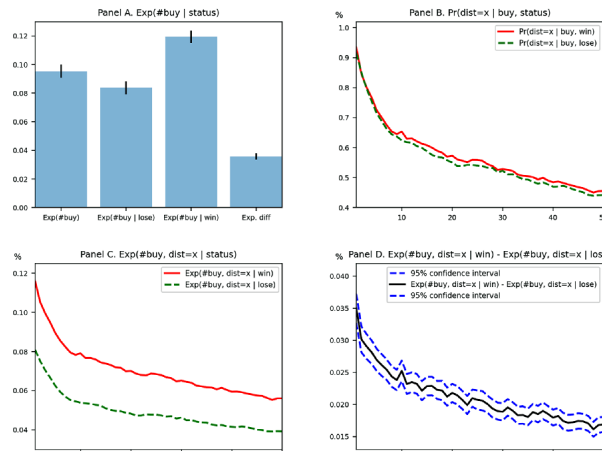
(b) Figure 2. Listing code and stock characteristics

6.2 Figure 3: Positive feedback trading and attention spillover and Table I: Positive Feedback Trading and Attention Spillover – Panel Regressions

Read: Figure 3 shows that investors buy more after winners than losers, buy nearer stocks more often, and that the win-minus-lose purchase difference declines with distance. Table I formalizes this in panel regressions with investor, stock, and date fixed effects; the winning-position coefficient is positive and strongest for close-distance bins.

Interpretation. Together, the figure and table verify both ingredients of the mechanism. Positive feedback trading alone would predict more buying after a winner, but not necessarily in neighbors. Attention spillover alone would predict proximity-based buying, but not stronger buying after winning experiences. The interaction is what generates stock-level buying pressure on neighbors of winners.

Economic message. The micro evidence makes the return tests credible: the stock-level signal is constructed from actual behavioral responses.



(a) Figure 3. Positive feedback trading and attention spillover

Table I
Positive Feedback Trading and Attention Spillover: Panel Regressions

This table shows how an investor's past performance affects their future purchases, as a function of the distance to the currently held stock. We randomly select 50,000 investors to form the sample and then perform a series of regressions of the form $I_{k,t+5} = \alpha_{k,t+5} I_{k,t} + \beta_{k,t+5} I_{k,t+1} + \text{controls} + \epsilon$. The dependent variable $I_{k,t+5}$ is a dummy indicating whether the investor buys any stocks whose distance to the currently held stock is between x and $x+5$, and the main independent variable of interest, $I_{k,t}$, indicates whether the current position has increased or decreased in value from the purchase date to the day in question. Panel A reports univariate results, while in Panel B we control for confounding factors $\text{Age}_{k,t+5}$ ($\text{Ind}_{k,t+5}$) indicates the fraction of stocks among the corresponding group (distance between x and $x+5$) that have the same listing year (industry) as the currently held stock. We include both $\text{Age}_{k,t+5}$ and $\text{Ind}_{k,t+5}$, as well as their interactions with $I_{k,t}$. All regressions in both panels include fixed effects for investor, date, and the currently held stock. Standard errors are clustered at the investor, date, and currently held stock level. All estimated parameters are reported in percentage, and t -statistics are reported in brackets.

	$I_{0,5}$	$I_{1,10}$	$I_{10,15}$	$I_{15,20}$	$I_{20,25}$	$I_{25,30}$	$I_{30,35}$	$I_{35,40}$	$I_{40,45}$	$I_{45,50}$
$I_{k,t}$	0.029	0.023	0.022	0.018	0.019	0.015	0.013	0.013	0.013	0.013
Intercept	115.191	114.061	115.000	113.161	114.831	112.161	110.801	110.631	110.811	110.181
Investor FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.004	0.004	0.004	0.003	0.003	0.003	0.003	0.003	0.003	0.003
No. of Obs.	85m	85m	85m	85m	85m	85m	85m	85m	85m	85m

(Continued)

(b) Table I, Panel A. Positive feedback regressions

Table I—Continued										
Panel B: Regressions controlling for age and industry effects										
$Y =$	$1_{0,0}$	$1_{0,10}$	$1_{10,10}$	$1_{10,20}$	$1_{20,20}$	$1_{20,30}$	$1_{30,30}$	$1_{30,40}$	$1_{40,40}$	$1_{40,50}$
1_{win}	0.046 [8.86]	0.031 [8.04]	0.025 [8.92]	0.017 [6.56]	0.018 [6.75]	0.008 [5.74]	0.005 [2.94]	0.007 [4.28]	0.009 [5.48]	0.007 [4.41]
$AB_{(i,t,t+1)}$	0.153 [3.66]	0.140 [4.59]	0.073 [3.64]	0.055 [4.90]	0.018 [3.03]	0.015 [1.77]	0.021 [2.73]	0.020 [3.14]	0.029 [5.14]	0.018 [3.04]
$Ind_{(i,t,t+1)}$	0.185 [2.30]	0.094 [2.76]	-0.017 [-0.77]	0.038 [1.46]	0.055 [2.53]	0.038 [2.54]	0.016 [1.25]	0.011 [1.04]	0.018 [1.60]	0.006 [0.59]
$1_{win} \times AB_{(i,t,t+1)}$	-0.035 [-5.00]	-0.020 [-3.71]	-0.013 [-2.82]	-0.002 [-0.39]	0.000 [0.06]	0.017 [3.90]	0.024 [5.06]	0.017 [3.77]	0.015 [3.38]	0.025 [4.45]
$1_{win} \times Ind_{(i,t,t+1)}$	0.123 [3.24]	0.086 [2.58]	0.083 [3.34]	0.065 [2.57]	0.031 [1.42]	0.048 [2.56]	0.035 [1.65]	0.049 [2.96]	0.035 [2.38]	0.060 [2.91]
<i>Intercept</i>	-0.054 [-1.27]	-0.014 [-0.88]	0.027 [2.98]	0.034 [6.91]	0.045 [19.79]	0.047 [16.19]	0.045 [19.24]	0.045 [26.28]	0.043 [32.99]	0.045 [40.17]
<i>Investor FE</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Stock FE</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Date FE</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.004	0.004	0.004	0.003	0.003	0.003	0.003	0.003	0.003	0.003
<i>No. of Obs.</i>	85m	85m	85m	85m	85m	85m	85m	85m	85m	85m

Table I, Panel B. Controls for age and industry effects

Table II
The Primary Determinants of Positive Feedback Trading

This table provides evidence on the factors that drive positive feedback trading. The sample contains 1,229 surveyed investors and their trading data from January 2020 to January 2022. Following Liu et al. (2022), the survey contains 12 questions designed to capture behavioral biases. We create a dummy variable for each bias, which we discuss in detail in Internet Appendix Section I.E. Panel A reports results of regressions of the form $1_{i,t+1}^{net\ buy} = \alpha_j + \beta_1 1_{i,t}^{bias_j} + \beta_2 1_{i,t}^{win} + \epsilon_{i,t+1}$, where the estimation of each regression is reported in one row. Panel B reports results of a horse race that includes all elicited biases in the regression $1_{i,t+1}^{net\ buy} = \alpha + \beta 1_{i,t}^{win} + \sum_j \beta_j 1_{i,t}^{bias_j} + \sum_j \gamma_j 1_{i,t}^{win} \times 1_{i,t}^{bias_j} + \epsilon_{i,t+1}$. Specifically, $1_{i,t+1}^{net\ buy}$ is a dummy variable indicating whether the i -th investor makes a net purchase on the $(t+1)$ -th day, $1_{i,t}^{win}$ is a dummy variable indicating whether the i -th investor currently has a winning or losing portfolio, and $1_{i,t}^{bias_j}$ is a dummy variable indicating the j -th behavioral bias for the i -th investor. All regressions include date fixed effects. All estimated parameters are reported in percentage, and t -statistics are reported in brackets.

Panel A. Regressions that contain a single behavioral bias						
	1_{win}	1_{bias}	$1_{win} 1_{bias}$	Date FE	R^2	No. of Obs.
Baseline	1.627 [5.23]			Yes	0.016	121,393
Overplacement	2.025 [4.28]	0.441 [1.46]	-0.675 [-1.09]	Yes	0.016	121,393
Upside self-attribution	0.818 [2.28]	0.211 [0.61]	2.915 [4.15]	Yes	0.016	121,393
Downside self-attribution	0.261 [0.71]	-1.683 [-5.20]	4.639 [6.84]	Yes	0.016	121,393
Upside extrapolation	1.251 [3.54]	-1.556 [-4.50]	1.631 [2.24]	Yes	0.016	121,393
Downside extrapolation	1.551 [4.50]	-0.638 [-1.76]	0.345 [0.44]	Yes	0.016	121,393
Perceived information advantage	1.080 [2.92]	-2.916 [-9.21]	1.809 [2.70]	Yes	0.016	121,393
Dismissiveness of others' information	1.299 [2.94]	0.001 [0.00]	0.629 [1.02]	Yes	0.016	121,393
Gambling preference, blockbusters	2.408 [6.45]	0.530 [1.70]	-2.567 [-3.87]	Yes	0.016	121,393
Gambling preference, lotteries	1.511 [4.01]	-1.122 [-3.64]	0.263 [0.40]	Yes	0.016	121,393
Realized utility, winner	1.484 [3.85]	1.893 [6.22]	0.473 [0.74]	Yes	0.016	121,393
Realized utility, loser	1.498 [3.68]	0.496 [1.64]	0.231 [0.37]	Yes	0.016	121,393
Home bias	1.376 [3.49]	0.226 [0.74]	0.632 [1.00]	Yes	0.016	121,393

(Continued)

Table II, Panel A. Single behavioral bias regressions

6.3 Table II: The Primary Determinants of Positive Feedback Trading

Read: Table II uses survey-linked trading data. Panel A evaluates each behavioral bias separately; Panel B runs a horse race across biases. Upside and downside self-attribution have the strongest interaction with winning status.

Interpretation. The table shows that positive feedback trading is not just extrapolation or gambling preference. It is most closely linked to self-attribution: investors appear to credit themselves for good outcomes and trade more afterward.

Economic message. The behavioral driver resembles the learning-to-be-overconfident mechanism in Gervais and Odean. This supports the paper’s interpretation of positive feedback trading as overconfidence-related.

table II—Continued

Panel B. A horse race of multiple behavioral biases		
	1_{bias}	$1_{win} 1_{bias}$
Overplacement	0.052 [0.17]	−0.015 [−0.02]
Upside self-attribution	0.592 [1.55]	2.198 [2.83]
Downside self-attribution	−1.960 [−5.55]	4.281 [5.74]
Upside extrapolation	−1.733 [−4.18]	1.638 [1.93]
Downside extrapolation	0.655 [1.48]	−1.608 [−1.76]
Perceived information advantage	−2.832 [−8.49]	1.332 [1.90]
Dismissiveness of others’ information	0.613 [2.02]	−1.548 [−0.24]
Gambling preference, blockbusters	0.996 [2.95]	−3.615 [−5.13]
Gambling preference, lotteries	−1.205 [−3.62]	0.408 [0.59]
Realized utility, winner	1.975 [6.42]	−0.189 [−0.29]
Realized utility, loser	0.811 [2.67]	−0.278 [−0.43]
Home bias	0.553 [1.73]	0.058 [0.09]

Table II, Panel B. Horse race of multiple behavioral biases

6.4 Table III: Experienced Returns versus Observed Extreme Returns

Read: Table III compares experienced returns to observed extreme returns around price-limit events. For extreme positive returns, holding the stock has a strong distance-decaying effect on neighbor purchases; the pure price-limit hit effect is much smaller. For extreme negative returns, effects are mostly insignificant.

Interpretation. The table separates “I experienced a gain” from “the stock was salient to everyone.” The holding effect dominates the price-limit hit effect, showing that personal experience matters much more than generic attention.

Economic message. This is a key mechanism falsification test: the stock-level pricing effect should come mainly from investors who actually benefited from neighboring stocks’ past returns.

Experienced Returns versus Observed Extreme Returns

This table compares the trading impact of experienced returns and observed extreme returns by analyzing events in which stocks experience extreme positive/negative returns. The sample is constructed as follows. Each day, we start with the full list of stocks hitting the upper (lower) daily price limit. For comparison, we also include the same number of stocks that almost hit the price limit on the same day, that is, those with extremely high (low) returns that did not reach the limit. We then construct the stock-day-investor sample as the Cartesian product of this daily stock list and 50,000 randomly selected investors who may or may not hold these stocks and run regressions of the form $y_{i,t+1} = \alpha_{i,t+1} + \beta' x_{i,t+1} + \epsilon_{i,t+1}$, where $y_{i,t+1}$ is the daily price limit at the end of the current day, and Panel B reports results for stocks experiencing extreme negative returns (hitting or almost hitting the lower price limit). We include fixed effects at the level of the investor, day, and focal stock. All estimated parameters are reported in percentage, and t -statistics are reported in brackets.

Panel A. The sample of stocks experiencing extreme positive returns (hitting or almost hitting the upper price limit)

$Y =$	$I_{10,10}$	$I_{10,10}$	$I_{10,10}$	$I_{10,10}$	$I_{10,10}$	$I_{10,10}$	$I_{10,10}$	$I_{10,10}$	$I_{10,10}$
$I_{10,10}$	0.022	0.018	0.013	0.013	0.004	0.003	0.007	0.006	0.003
I_{10}	[5.67]	[3.55]	[2.72]	[2.73]	[1.55]	[1.44]	[0.79]	[0.79]	[1.91]
$I_{10} \times I_{10}$	0.003	0.002	0.001	0.003	0.001	0.002	0.003	0.002	0.001
	[1.58]	[1.26]	[0.67]	[1.36]	[0.99]	[1.58]	[1.83]	[1.21]	[1.46]
$I_{10} \times I_{10}$	0.008	0.012	0.011	0.005	0.014	0.005	0.017	0.006	0.011
	[1.62]	[1.48]	[1.47]	[0.78]	[2.04]	[0.83]	[2.44]	[0.87]	[1.70]
$Intercept$	0.089	0.088	0.088	0.086	0.087	0.085	0.085	0.085	0.085
	[205.79]	[233.68]	[193.21]	[194.84]	[205.02]	[209.01]	[160.03]	[231.07]	[268.87]
Investor FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.003

Table III, Panel A. Extreme positive returns

Table III—Continued

Panel B. The sample of stocks experiencing extreme negative returns (hitting or almost hitting the lower price limit)

$Y =$	$I_{10,10}$	$I_{10,10}$	$I_{10,10}$	$I_{10,10}$	$I_{10,10}$	$I_{10,10}$	$I_{10,10}$	$I_{10,10}$	$I_{10,10}$
$I_{10,10}$	0.016	0.012	0.001	0.002	0.003	0.006	-0.002	0.004	0.007
I_{10}	[1.72]	[1.60]	[0.23]	[0.21]	[0.36]	[0.88]	[-0.34]	[0.58]	[0.98]
$I_{10} \times I_{10}$	-0.001	0.000	-0.002	-0.001	0.000	0.001	0.002	0.000	-0.001
	[-0.71]	[-0.07]	[-1.80]	[-0.61]	[-0.32]	[1.20]	[1.48]	[-0.02]	[-0.98]
$I_{10,10} \times I_{10}$	-0.001	-0.003	-0.008	0.016	0.020	0.001	0.006	-0.009	0.011
	[-0.08]	[-0.22]	[-0.84]	[1.33]	[1.34]	[0.10]	[0.63]	[-1.05]	[1.08]
$Intercept$	0.031	0.031	0.031	0.030	0.030	0.029	0.029	0.031	0.030
	[71.64]	[89.91]	[119.10]	[65.70]	[80.30]	[86.01]	[86.84]	[80.56]	[145.55]
Investor FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.005	0.005	0.005	0.005	0.005	0.005	0.005	0.005	0.005
No. of Obs.	24m	24m	24m	24m	24m	24m	24m	24m	24m

Table III, Panel B. Extreme negative returns

6.5 Table IV: Summary Statistics and Table V: Single-Sorted Portfolio Return by RLOCAL

Read: Table IV shows that RLOCAL portfolios are broadly balanced on most firm characteristics; correlations between RLOCAL and controls are close to zero. Table V shows monotonic next-week returns across RLOCAL quintiles and a long-short spread of roughly 8% per year across weighting schemes.

Interpretation. Table IV supports the identification premise that RLOCAL is not simply a disguised size, beta, book-to-market, or momentum variable. Table V establishes the core pricing fact: stocks adjacent to recent winners subsequently outperform stocks adjacent to recent losers.

Economic message. The result is consistent with temporary attention-induced demand: RLOCAL is designed to isolate neighbor-performance-induced buying pressure.

Table IV
Summary Statistics

This table reports summary statistics for key variables and control variables in our asset pricing tests. We report the time-series average of the equal-weighted stock characteristics for portfolios sorted by RLOCAL and the time-series average of the cross-sectional correlation between characteristics and RLOCAL (LOCAL). For each stock, LOCAL is the value-weighted average return over the past two weeks of the 10 closest stocks on the screen display. RLOCAL is the residual of the cross-sectional regression of LOCAL on the focal stock's return over the past two weeks. $Ret_{i,t-12:t-1}$, $Ret_{i,t-12:t-1}$, and $Ret_{i,t-12:t-1}$ denote the cumulative return of the own stock in the past two weeks, from month $t-12$ to month $t-1$, and from month $t-12$ to month $t-1$, respectively. $LogAge$ is the logarithm of the number of months since the firm's IPO. Beta is estimated using the CAPM and the monthly returns in the past 36 months. $LogME$ is the logarithm of a firm's market capitalization at the end of the week. $LogBM$ is the logarithm of the book-to-market ratio. $ILLIQ$ denotes the Amihud illiquidity measure, calculated as the average daily ratio of the absolute return over trading volume in the past four weeks. $Turnover$ is the average daily ratio of trading volume over total tradable shares in the past four weeks (in percentage points). $IVOL$ is idiosyncratic volatility with respect to the Fama-French three-factor model using daily returns over the past four weeks. Max is the average of the three highest daily returns over the past four weeks. $Skew$ is skewness of daily returns over the past 52 weeks. All variables are winsorized each week at the 1% and 99% levels. t -Statistics are shown in brackets.

	RLOCAL	LOCAL	Ret _{i,t-12:t-1}	LogAge	LogME	Beta	LogBM	Ret _{i,t-12:t-1}	ILLIQ	Turnover	IVOL	Max	Skew
RLOCAL1	-0.042	-0.036	0.006	4.517	22.197	1.101	-1.055	0.191	0.507	0.186	2.292	0.019	0.055
RLOCAL2	-0.017	-0.012	0.005	4.574	22.163	1.101	-1.050	0.185	0.510	0.186	2.272	0.019	0.055
RLOCAL3	-0.002	0.003	0.005	4.572	22.156	1.102	-1.048	0.187	0.509	0.186	2.292	0.019	0.055
RLOCAL4	0.014	0.020	0.005	4.551	22.149	1.100	-1.053	0.188	0.509	0.187	2.308	0.019	0.055
RLOCAL5	0.048	0.051	0.006	4.493	22.176	1.097	-1.053	0.194	0.507	0.184	2.349	0.019	0.055
RLOCAL5-1	0.087	0.087	0.000	-0.024	-0.021	-0.004	0.002	0.003	0.000	-0.002	0.007	0.000	-0.007
	[25.35]	[25.36]	[1.22]	[-1.15]	[-1.39]	[-1.58]	[0.22]	[0.78]	[0.03]	[-1.28]	[2.33]	[-1.03]	[-0.78]
Corr. RLOCAL	1.00	1.00	0.00	-0.02	-0.01	0.00	0.00	0.00	0.00	0.01	0.00	0.00	-0.01
Corr. LOCAL	1.00	1.00	0.06	-0.02	0.00	0.00	0.00	0.00	0.00	0.02	0.01	0.02	0.00

Table IV. Summary statistics by RLOCAL

Table V
Single-Sorted Portfolio Return by RLOCAL

This table reports single-sorted portfolio results. At the end of each week, we sort stocks into five groups based on RLOCAL and we track the returns of these portfolios over the next week. For each stock, LOCAL is the value-weighted average return over the past two weeks of the 10 closest stocks on the screen display, and RLOCAL is the residual of the cross-sectional regression of LOCAL on the focal stock's return over the past two weeks. We report portfolio returns using three weighting schemes: equal weighting (EW), value weighting (VW), and capped-value weighting (CVW), which weights stocks by their market value winsorized at the 80th percentile across all stocks following Jensen, Kelly, and Pedersen (2021). We also report the long-short portfolio return (P5-P1) as well as a battery of risk-adjusted returns using different benchmarks: age-adjusted return (Age-adj Ret), industry-adjusted return (Ind-adj Ret), DGTW characteristic-adjusted return (DGTW Ret), four-factor alpha (CH4 Alpha) following Lim, Stambaugh, and Yuan (2019), and five-factor alpha (FF5 Alpha) following Fama and French (2015). All returns and alphas are annualized and reported in percentage points. The sample is from January 2002 to December 2019. t -Statistics, shown in brackets, are based on standard errors with Newey-West adjustments of 12 lags.

	P1	P2	P3	P4	P5	P5-P1	Age-adj Ret	Ind-adj Ret	DGTW Ret	CH4 Alpha	FF5 Alpha
EW	8.120	10.942	12.401	13.274	16.140	8.020	7.229	5.625	4.146	8.929	7.979
	[0.88]	[1.18]	[1.34]	[1.41]	[1.71]	[5.45]	[5.51]	[5.45]	[3.94]	[5.51]	[5.45]
VW	3.311	6.557	8.715	8.628	11.822	8.511	7.427	2.995	3.827	11.825	7.862
	[0.46]	[0.80]	[1.06]	[1.00]	[1.43]	[2.67]	[3.26]	[2.65]	[2.08]	[0.20]	[2.51]
CVW	5.471	8.293	9.265	10.050	12.367	6.896	6.304	4.278	3.173	7.697	6.933
	[0.61]	[0.92]	[1.03]	[1.10]	[1.36]	[4.59]	[4.64]	[4.39]	[2.84]	[4.54]	[4.61]

Table V. Single-sorted portfolio return by RLOCAL

6.6 Table VI: Double-Sorted Portfolio Return by RLOCAL and Confounding Variables and Table VII: Fama-MacBeth Regressions

Read: Table VI double-sorts on RLOCAL within characteristic groups such as size, beta, book-to-market, past returns, illiquidity, turnover, volatility, maximum return, skewness, and exchange. Table VII estimates Fama-MacBeth regressions of future returns on LOCAL and controls.

Interpretation. The return spread survives controls for standard return predictors. This matters because a neighbor-return variable could otherwise proxy for omitted characteristics or industry/age effects.

Economic message. The return predictability is not just a repackaged anomaly. It is tied to the screen-adjacency attention channel.

Table VI
Double-Sorted Portfolio Return by RLOCAL and Confounding Factors

This table reports double-sorted portfolio results based on *RLOCAL* and various control variables, including beta, firm size (*logME*), book-to-market ratio (*logBM*), illiquidity (*ILLIQ*), turnover, idiosyncratic volatility (*IVOL*), max return (*Max*), return skewness (*Skew*), past returns over different horizons, and the stock's listing board. At the end of each week, for each control variable we sort stocks into five quintiles based on this variable, within each quintile, we divide stocks into five groups based on *RLOCAL* and we collapse across the groups based on the control variable. We report the long-short portfolio return (P5-P1) as well as a battery of risk-adjusted returns using different benchmarks: age-adjusted return (*Age-adj Ret*), industry-adjusted return (*Ind-adj Ret*), DGTW characteristic-adjusted return (*DGTW Ret*), four-factor alpha (*CH4 Alpha*), and five-factor alpha (*FF5 Alpha*). All returns and alphas are annualized and reported in percentage points. *t*-Statistics, shown in brackets, are based on standard errors with Newey-West adjustments of 12 lags.

	LogME		Beta		LogBM		Ret _{-12m, -2m}		Ret _{-36m, -13m}		ILLIQ
	EW	VW	EW	VW	EW	VW	EW	VW	EW	VW	EW
P5-P1	6.312	5.908	7.877	6.806	6.871	5.649	7.878	7.915	7.148	7.146	6.007
	[4.96]	[4.30]	[5.54]	[2.89]	[5.35]	[2.26]	[5.61]	[3.16]	[5.11]	[2.88]	[4.58]
Age-adj Ret	5.698	5.506	7.080	6.244	6.181	5.046	7.221	7.328	6.515	6.307	5.422
	[4.95]	[4.62]	[5.51]	[3.37]	[5.24]	[2.60]	[5.74]	[3.85]	[4.99]	[3.24]	[4.55]
Ind-adj Ret	4.516	3.861	5.557	2.852	5.169	2.735	5.598	3.962	5.310	3.289	4.193
	[4.59]	[3.97]	[5.38]	[2.61]	[5.36]	[2.33]	[5.67]	[3.89]	[5.29]	[2.94]	[4.36]
DGTW Ret	4.241	4.220	4.379	3.183	4.074	2.246	4.183	3.703	3.916	3.780	3.730
	[3.83]	[3.63]	[4.14]	[2.13]	[4.02]	[1.35]	[4.11]	[2.56]	[3.83]	[2.39]	[3.50]
CH4 Alpha	6.857	6.728	8.739	8.032	7.478	7.864	8.512	9.762	7.825	8.703	6.839
	[4.78]	[4.25]	[5.52]	[3.05]	[5.35]	[2.86]	[5.72]	[3.48]	[5.21]	[3.08]	[4.63]
FF5 Alpha	6.328	5.882	7.791	6.422	6.814	4.901	7.743	7.193	6.976	6.202	5.956
	[5.05]	[4.33]	[5.42]	[2.73]	[5.37]	[1.99]	[5.68]	[3.08]	[5.26]	[2.60]	[4.53]
	ILLIQ		Turnover		IVOL		Max		Skew		Board
	VW	EW	VW	EW	VW	EW	VW	EW	VW	EW	VW
P5-P1	5.660	7.481	7.516	8.031	8.477	7.459	8.270	7.681	8.733	4.810	3.727

(a) Table VI. Double-sorted portfolio returns

Table VII
Fama-MacBeth Regressions

This table reports Fama-MacBeth regression results where the dependent variable is the future one-week stock return (in percentage). The main independent variable of interest, *LOCAL*, is calculated as the value-weighted average return over the past two weeks of the 10 stocks that are closest in listing code to the focal stock. We also control for various stock characteristics, including beta, firm size (*logME*), book-to-market ratio (*logBM*), illiquidity (*ILLIQ*), turnover, idiosyncratic volatility (*IVOL*), max return (*Max*), return skewness (*Skew*), past returns over different horizons, and a number of dummy variables indicating the industry and listed year. All of the independent variables except those related to returns are winsorized each week at the 1% and 99% levels. *t*-Statistics, shown in brackets, are based on standard errors with Newey-West adjustments of 12 lags.

	[1]	[2]	[3]	[4]	[5]	[6]
<i>LOCAL</i>	0.823	0.667	0.801	0.422	0.611	0.380
	[3.90]	[3.55]	[3.71]	[2.30]	[3.43]	[2.23]
<i>Ret_{-2w}</i>		-2.889		-2.587		-2.939
		[-10.04]		[-8.50]		[-10.05]
<i>LogME</i>		-0.055		-0.052		-0.058
		[-1.55]		[-1.37]		[-1.61]
<i>Beta</i>		0.052		0.061		0.059
		[1.22]		[1.13]		[1.39]
<i>LogBM</i>		0.036		0.019		0.033
		[1.51]		[0.66]		[1.40]
<i>Ret_{-12m, -2m}</i>		0.149		0.167		0.145
		[1.95]		[2.05]		[1.89]
<i>Ret_{-36m, -13m}</i>		-0.023		-0.027		-0.025
		[-0.72]		[-0.84]		[-0.86]
<i>ILLIQ</i>		4.194		4.015		3.991
		[5.59]		[5.21]		[5.36]
<i>Turnover</i>		-8.183		-7.924		-8.613
		[-5.90]		[-5.65]		[-6.34]
<i>IVOL</i>		-26.228		-25.993		-25.513
		[-13.35]		[-12.80]		[-12.87]
<i>Max</i>		5.259		5.893		5.118
		[7.18]		[6.10]		[7.03]
<i>Skew</i>		-0.002		0.005		-0.000
		[-0.04]		[0.12]		[-0.01]
Ind FE	Yes	Yes	No	No	Yes	Yes
Age FE	No	No	Yes	Yes	Yes	Yes
Avg. weekly obs.	1,616	1,566	1,616	1,566	1,616	1,566
Adj-R ²	0.071	0.141	0.012	0.100	0.076	0.145

(b) Table VII. Fama-MacBeth regressions

6.7 Table VIII: Fama-MacBeth Regressions for Placebo Tests and Table IX: Forecasting Turnover by RLOCAL

Read: Table VIII uses placebo LOCAL variables and distant neighbors. Table IX shows that LOCAL predicts future turnover and small-order buying-related turnover.

Interpretation. The placebo tests show that return predictability is strongest where attention spillover should operate. The turnover evidence closes the loop: the price effect is accompanied by trading activity, especially consistent with retail buy pressure.

Economic message. If LOCAL predicted returns but not turnover, the mechanism would be weaker. The turnover result supports a demand-pressure interpretation.

6.8 Figure 4: Comovement in Return and Turnover and Distance between Stocks and Table X: Comovement in Return and Turnover Difference-in-Difference

Read: Figure 4 shows that return and turnover correlations are higher for close listing-code neighbors and decline with distance, while accounting fundamentals do not show the same distance pattern. Table X uses the SME Board display-order change and finds lower comovement when pairs become farther apart.

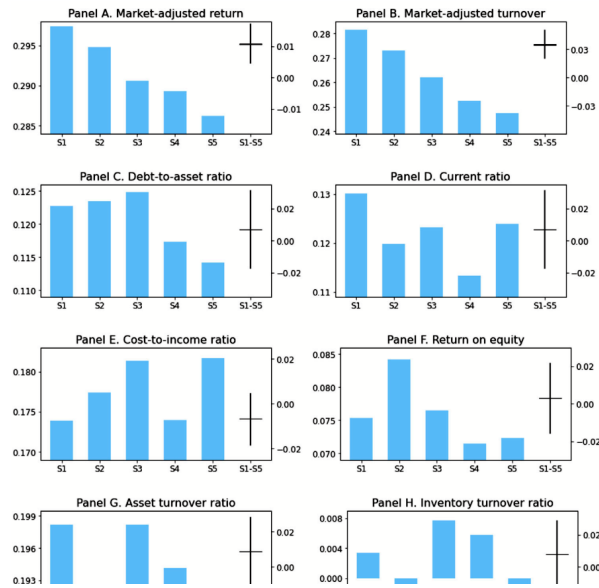
Interpretation. Figure 4 documents the cross-sectional comovement implication of attention spillover; Table X strengthens the causal interpretation by using a shock to display distance.

This table reports results of Pama-Belt regressions in which the dependent variable is the future one-week stock return (in percentage). The main independent variable of interest, *LOCAL*, is the value-weighted average return over the past two weeks of the 10 stocks that are closest in geographic location to the stock under consideration. To construct *LOCAL*, we first identify the ten nearest stocks from the NYSE-AMEX database, then we select the ten stocks with the closest listing dates and construct the placebo variable using returns of the next 10 stocks. Second, we replace returns of neighboring stocks with the turnover of these stocks over the past two weeks. The third construction uses return volatility of neighboring stocks instead of returns. The regression results across these three placebo variables are reported in Panels A, B, and C, respectively. We also control for a number of other factors: size (beta), firm age (AGE), leverage (LEV), industry fixed effects (INDUSTRY), and market capitalization (MCAP). Panel D shows the maximum return (*Max*) return as a benchmark, past returns over different horizons, and dummy variables indicating whether the stock is listed year. All of the independent variables except those related to returns are winsorized each day at the 1% and 99% levels. *t*-Statistics shown in brackets, are based on standard errors with Newey-West adjustments of 12 lags.

[illegible]

Table VIII. Fama-MacBeth regressions for placebo tests

Economic message. The evidence suggests that trading-platform design can create non-fundamental comovement among stocks.



(a) Figure 4. Comovement and display distance

Table IX
Forecasting Turnover by *RLOCAL*

This table reports the average daily turnover and order imbalance over the next week for single-sectored portfolios based on *RLLOCAL*. At the end of each week, we sort stocks into five groups based on *RLLOCAL* and track the average daily turnover and small order imbalance of these portfolios over the next week. For each stock, *LOCAL* is the value-weighted average returns over the past two weeks of the 10 closest stocks on screen display, and *RLLOCAL* is the raw value-weighted, and capped-value-weighted turnover, as well as abnormal turnover, which is the difference between the average weekly turnover and the average of this value over the past 52 weeks. We also report results for small order imbalance, calculated as the difference between buyer-initiated small trades and seller-initiated small trades, normalized by total trades each day, and then take the average over the week. Small trades are those with a value less than \$100,000. The sample is *LOCAL* January 2002 to December 2019 for the turnover variables, and from January 2003 to December 2019 for order imbalance variables. Daily values are reported as a percentage. *t*-Statistics, shown in brackets, are based on standard errors with Newey-West adjustments of 12 lags.

	Turnover			Annual Turnover			Small OIB NYL			Small OIB VOL		
	EW	VM	CVW	EW	VM	CVW	EW	VM	CVW	EW	VM	CVW
P1	2.295	1.500	2.029	-0.075	-0.073	-0.067	-0.611	-0.393	-0.438	-0.610	-0.393	-0.438
P2	2.295	1.526	2.016	-0.074	-0.074	-0.054	-0.614	-0.318	-0.419	-0.634	-0.318	-0.419
P3	2.284	1.565	2.024	-0.062	-0.045	-0.053	-0.614	-0.302	-0.442	-0.614	-0.302	-0.442
P4	2.313	1.601	2.041	-0.051	-0.032	-0.047	-0.600	-0.270	-0.410	-0.600	-0.271	-0.413
P5	2.372	1.602	2.065	-0.048	-0.048	-0.029	-0.505	-0.275	-0.400	-0.580	-0.275	-0.400
P5-P1	0.076	0.076	0.036	0.073	0.075	0.038	0.098	0.038	0.038	0.027	0.038	0.038
	[2.83]	[2.64]	[2.09]	[4.09]	[4.33]	[2.77]	[1.90]	[3.22]	[2.56]	[1.90]	[1.30]	[2.56]

Table IX. Forecasting turnover by RLOCAL

Comovement in Return and Turnover: Difference-in-Difference (DID) Approach

This table reports the return and turnover correlations between treatment stocks and control stocks before and after the introduction of the SME Board in May 2004. Panel A reports results on return correlations, and Panel B reports results on turnover correlations. The before period is March to May, while the after period is October to December. 000Z and 000Y indicate the last 20 stocks and the second-last 20 stocks listed on the Shenzhen Main Board (where listing codes start with 000), respectively, while 600A and 600B refer to the first 20 stocks and the 20th to 40th stocks listed on the Shanghai Main Board (where listing codes start with 600). Our sample includes stocks that have at least 15 observations in both before and after periods. $\rho(000Y, 000Z)$ is the average pairwise Spearman-rank correlation of daily returns or turnover between stocks in the 000Y group and stocks in the 000Z group; likewise for other groups. "Diff" denotes the difference between stock groups or time periods. *t*-Statistics are shown in brackets.

Panel A. DID tests on return correlation						
	$\rho(000Y, 000Z)$	$\rho(000Z, 600A)$	Diff	$\rho(000Z, 600A)$	$\rho(600A, 600B)$	Diff
Before	0.415 [55.62]	0.382 [60.85]	-0.033 [-3.35]	0.382 [60.85]	0.384 [62.26]	0.002 [0.24]
After	0.415 [53.87]	0.302 [36.64]	-0.113 [-9.59]	0.302 [36.64]	0.333 [43.90]	0.032 [2.85]
Diff	0.000 [-0.04]	-0.080 [-8.38]	-0.080 [-6.09]	-0.080 [-8.38]	-0.051 [-6.42]	0.030 [2.55]
No.	400	400	400	400	400	400

Panel B. DID tests on turnover correlation						
	$\rho(000Y, 000Z)$	$\rho(000Z, 600A)$	Diff	$\rho(000Z, 600A)$	$\rho(600A, 600B)$	Diff
Before	0.415 [55.62]	0.382 [60.85]	-0.033 [-3.35]	0.382 [60.85]	0.384 [62.26]	0.002 [0.24]
After	0.415 [53.87]	0.302 [36.64]	-0.113 [-9.59]	0.302 [36.64]	0.333 [43.90]	0.032 [2.85]
Diff	0.000 [-0.04]	-0.080 [-8.38]	-0.080 [-6.09]	-0.080 [-8.38]	-0.051 [-6.42]	0.030 [2.55]
No.	400	400	400	400	400	400

(b) Table X. Difference-in-difference comovement test

6.9 Table XI: Performance of RLOCAL Portfolios in the Post-SME Period and Figure 5: Cumulative Return of the Long-Short Portfolio Based on RLOCAL

Read: Table XI shows RLOCAL portfolio performance after the SME Board introduction. Figure 5 shows that the long-short CH4 alpha rises in the short run but reverses and vanishes within about 10 to 18 weeks, depending on weighting.

Interpretation. The post-SME evidence confirms that return predictability remains visible in the relevant period. The reversal result is crucial: if RLOCAL reflected fundamental information, returns should not fully unwind.

Economic message. The pattern looks like temporary price pressure from attention-driven buying, not a permanent change in expected cash flows.

Table XI
Performance of RLOCAL Portfolio in Pre-2002 Period

This table reports results of a single sort on *RLOCAL* from 1996 to 2001. We start the sample in 1996 to ensure that each portfolio has at least 30 stocks. For each stock, *LOCAL* is the value-weighted average return over the past two weeks of the 10 closest stocks on the screen display, and *RLOCAL* is the residual of the cross-sectional regression of *LOCAL* on the focal stock's return over the past two weeks. Raw and adjusted returns of hedge portfolios are reported. All returns and alphas are annualized and reported in percentage points. *t*-Statistics, shown in brackets, are based on standard errors with Newey-West adjustments of 12 lags.

	P1	P2	P3	P4	P5	P5-P1	Age-adj Ret	Ind-adj Ret	DGTW Ret	CH4 Alpha	FF5 Alpha
EW	28.059 [2.19]	29.977 [2.15]	26.186 [2.01]	30.088 [2.21]	30.799 [2.12]	2.140 [0.42]	2.159 [0.44]	0.938 [0.20]	-0.085 [-0.02]	1.249 [0.24]	-0.075 [-0.02]
VW	24.560 [1.72]	25.347 [1.75]	24.704 [1.67]	27.708 [1.76]	24.837 [1.62]	0.276 [0.05]	-0.117 [-0.00]	-0.023 [-0.00]	0.032 [0.01]	0.115 [0.02]	-3.037 [-0.63]
CWV	25.385 [1.86]	26.839 [1.90]	24.688 [1.73]	27.387 [1.94]	27.054 [1.77]	1.769 [0.32]	1.654 [0.32]	0.917 [0.18]	0.322 [0.24]	0.116 [0.02]	-0.664 [-0.13]

(a) Table XI. Post-SME RLOCAL portfolio performance

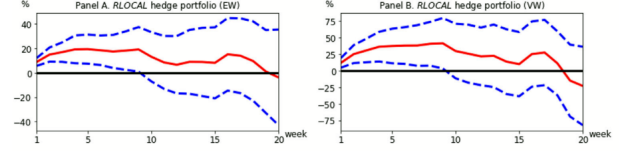


Figure 5. Cumulative return of the long-short portfolio based on *RLOCAL*. This figure shows the annualized cumulative CH4-alpha (in percentage points) to the equal- and value-weighted long-short portfolios (P5-P1) based on *RLOCAL* from week $t + 1$ to week $t + 20$, as well

(b) Figure 5. Cumulative return of RLOCAL long-short portfolio

6.10 Table XII: Cost of Arbitrage and RLOCAL Returns

Read: Table XII sorts RLOCAL portfolios within high and low arbitrage-cost groups. The RLOCAL spread is stronger among smaller, less liquid, and less-covered stocks.

Interpretation. Behavioral price pressure should be corrected more slowly when arbitrage is difficult. The results follow that prediction: the anomaly is stronger where limits to arbitrage are stronger.

Economic message. This links the behavioral mechanism to asset-pricing limits. Attention-driven demand can move prices more when sophisticated capital is less able or less willing to offset the pressure.

Table XII
Heterogeneous Tests

This table reports double-sorted portfolio results based on *RLOCAL* and various proxies for arbitrage costs, including firm size, illiquidity, and analyst coverage. At the end of each week, we sort stocks into two groups based on each of the moderating factors, and then sort stocks into five quintiles based on *RLOCAL* within each group. We report the value-weighted returns over the next week for the long-short portfolio (P5-P1) in high- and low-moderator groups. All returns and alphas are annualized and reported in percentage points. *t*-Statistics, shown in brackets, are based on standard errors with Newey-West adjustments of 12 lags.

		P5-P1		Age-adj Ret		Ind-adj Ret		DGTW Ret		CH4 Alpha		FF5 Alpha	
		para	t	para	t	para	t	para	t	para	t	para	t
LogME	L	9.177	[5.29]	9.217	[5.32]	7.691	[5.16]	8.239	[4.95]	9.854	[5.30]	9.300	[5.65]
	H	5.737	[1.82]	4.719	[2.14]	0.902	[0.75]	2.581	[1.22]	8.323	[2.22]	4.846	[1.57]
ILLIQ	L	5.824	[1.88]	4.751	[2.23]	0.733	[0.60]	2.207	[1.07]	8.883	[2.43]	4.820	[1.58]
	H	8.465	[5.02]	7.919	[4.83]	6.333	[4.61]	5.890	[3.73]	8.001	[4.55]	8.511	[5.17]
Analyst	L	12.795	[5.13]	12.187	[5.25]	9.171	[4.66]	7.758	[3.59]	14.103	[5.08]	13.011	[4.96]
	H	5.654	[1.22]	4.193	[1.31]	-1.341	[-0.79]	1.862	[0.62]	9.166	[1.78]	4.718	[1.08]

Table XII. Cost of arbitrage and RLOCAL returns

7 Interpretive synthesis

- **Mechanism chain.** Personal gain → overconfidence/self-attribution → more trading → screen-neighbor attention → buying pressure → short-term return and turnover effects → reversal.
- **Why this is not standard momentum.** The predictor is not the focal stock's return but the return of screen neighbors, and it is residualized from the focal stock's own past return.
- **Why this is not fundamentals.** Nearby listing-code stocks do not show parallel distance gradients in accounting fundamentals, and return effects reverse.
- **Why this is not attention alone.** Pure attention proxies such as price-limit hits, turnover, and volatility do not generate comparable results once the experienced-return mechanism is accounted for.

- **Why it matters for market design.** Interface design can create correlated attention and non-fundamental comovement. Trading platforms are not neutral display technologies.

8 Short summary

- The paper studies how overconfidence and limited attention interact in asset pricing.
- The empirical lever is listing-code-based stock display in Chinese trading software.
- Retail investors buy more after holding winners than after holding losers.
- Investors' new purchases are more likely to be close in display distance to their current holdings.
- Survey evidence points to self-attribution as a main driver of positive-feedback trading.
- LOCAL measures recent performance of screen-neighbor stocks; RLOCAL removes focal-stock past-return effects.
- High RLOCAL predicts higher next-week returns and turnover.
- The return spread survives characteristic controls, risk adjustments, placebo tests, and Fama-MacBeth regressions.
- Comovement in returns and turnover declines with listing-code distance; accounting fundamentals do not show the same pattern.
- A display-distance shock from the SME Board introduction lowers comovement for affected pairs.
- RLOCAL returns reverse over time, consistent with temporary price pressure.
- The effect is stronger among stocks with higher limits to arbitrage.

9 Conclusion

9.1 Core conclusion

Chen, An, Wang, and Yu show that overconfidence and limited attention interact to create temporary asset-pricing effects. Investors who experience gains trade more, and because they pay attention to screen-neighboring stocks, their trading pressure spills over to nearby stocks in display order.

9.2 Economic intuition

A gain in a currently held stock changes the investor's willingness to trade. The trading screen then shapes which stocks are considered. Neighboring stocks become salient precisely to investors who have just become more willing to buy. This creates short-term demand pressure that raises prices and turnover but eventually reverses.

9.3 Incremental contribution revisited

The paper's contribution is not simply another attention anomaly. It identifies a precise interaction: attention determines where demand pressure is directed, while positive-feedback trading determines when demand pressure rises. The combination explains why irrelevant display adjacency can forecast future returns.

9.4 What the paper does not claim

The paper does not claim that all attention effects are driven by trading screens, that all investors are overconfident, or that listing-code proximity always dominates fundamentals. Rather, it shows that a clean, exogenous source of display proximity can reveal a broader behavioral channel.

9.5 Final takeaway

The paper turns a small institutional detail—the order in which stocks appear on a trading screen—into a powerful test of behavioral asset pricing. Its key lesson is that market prices can be shaped by the interaction between investor psychology and market-interface design.